

Corrigé — Exercice 8

1) *Init.* : $0 < u_0 = 1 < 3$. *Héréd.* : si $0 < u_n < 3$, alors $u_{n+1} = \frac{4u_n}{1+u_n} > 0$; et $u_{n+1} < 3 \Leftrightarrow 4u_n < 3(1+u_n) \Leftrightarrow u_n < 3$. Donc $0 < u_{n+1} < 3$.

2a) $u_{n+1} - u_n = \frac{4u_n - u_n(1+u_n)}{1+u_n} = \frac{u_n(3-u_n)}{1+u_n} > 0$ (car $0 < u_n < 3$) : croissante.

2b) Croissante et majorée par 3, donc convergente vers ℓ ; $\ell = \frac{4\ell}{1+\ell} \Rightarrow \ell^2 = 3\ell \Rightarrow \ell \in \{0, 3\}$.
Comme $u_n \geq u_0 = 1$, $\ell = 3$.

3a) $u_{n+1} - 3 = \frac{4u_n - 3(1+u_n)}{1+u_n} = \frac{u_n - 3}{1+u_n}$, donc $v_{n+1} = \frac{u_{n+1} - 3}{u_{n+1}} = \frac{(u_n - 3)/(1+u_n)}{4u_n/(1+u_n)} = \frac{u_n - 3}{4u_n} = \frac{1}{4}v_n$: géométrique de raison $\frac{1}{4}$, $v_0 = \frac{1-3}{1} = -2$.

3b) $v_n = -2\left(\frac{1}{4}\right)^n \rightarrow 0$.

4a) $v_k = \frac{u_k - 3}{u_k} = 1 - \frac{3}{u_k} = 1 - w_k$, donc $w_k = 1 - v_k$.

4b) $S_n = \sum_{k=0}^n (1 - v_k) = (n+1) - \sum_{k=0}^n v_k = (n+1) + 2 \cdot \frac{4}{3} \left(1 - \left(\frac{1}{4}\right)^{n+1}\right) = n+1 + \frac{8}{3} \left(1 - \left(\frac{1}{4}\right)^{n+1}\right)$.

4c) $\frac{S_n}{n} = \frac{n+1}{n} + \frac{1}{n} \cdot \frac{8}{3} \left(1 - \left(\frac{1}{4}\right)^{n+1}\right) \rightarrow 1$.